



SIMULIA流体解决方案

Mar 19th, 2025



3DEXPERIENCE®

沈旭 Grace.SHEN@3ds.com
达索系统SIMULIA流体技术顾问

AGENDA

SIMULIA流体产品总览

PowerFLOW

XFlow、FMK、VPML

2025合作伙伴培训计划

AGENDA

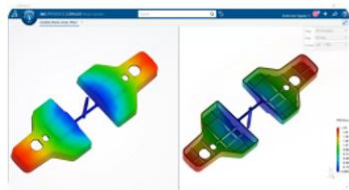
SIMULIA流体产品总览

PowerFLOW

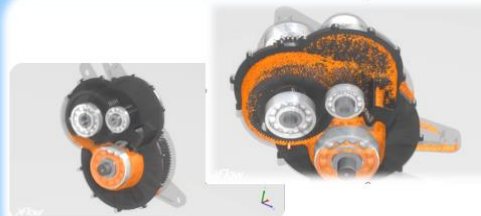
XFlow、FMK、VPML

2025合作伙伴培训计划

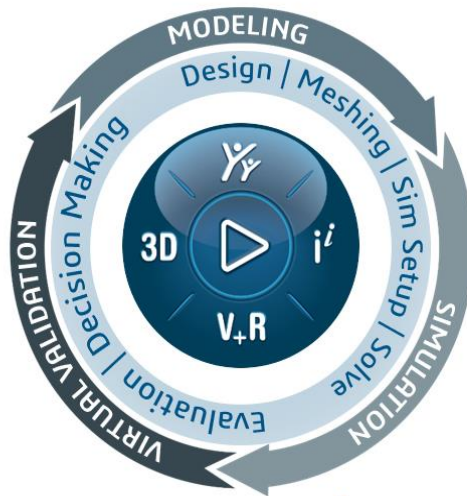
Plastic Injection Engineer



XFlow

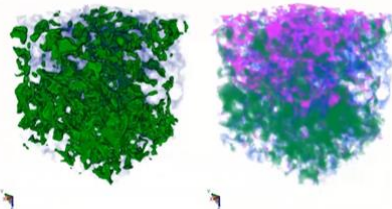


MODSIM

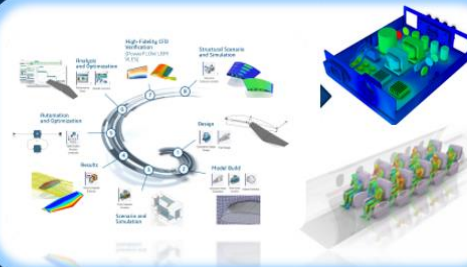


Porous Media Characterization Engineer

PV=0.19



Fluid Dynamics Engineer



PowerFLOW



AGENDA

SIMULIA流体产品总览

PowerFLOW

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2025合作伙伴培训计划

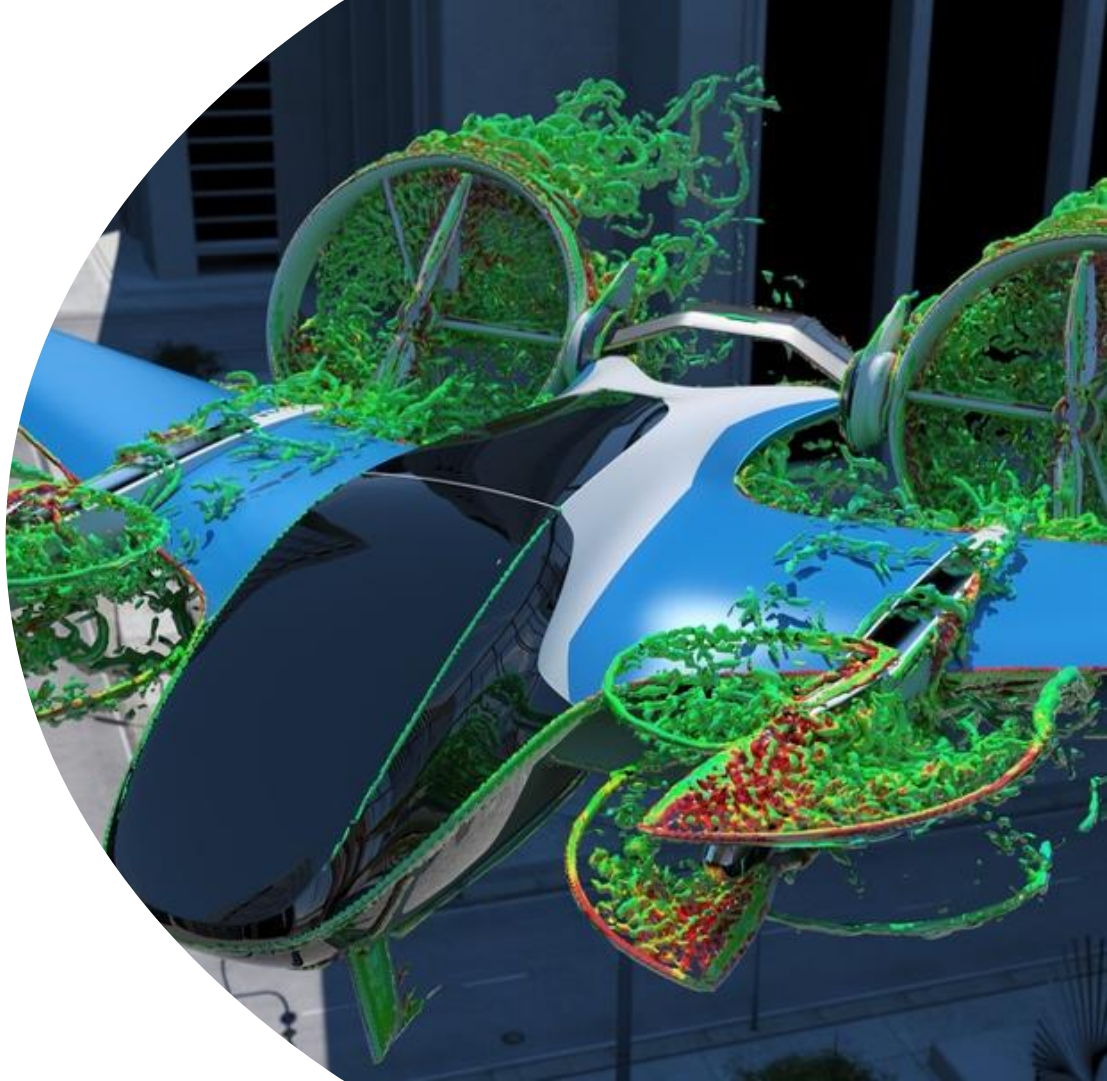


POWERFLOW

风噪仿真专家



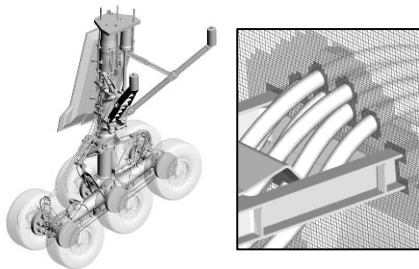
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PowerFLOW求解器优势

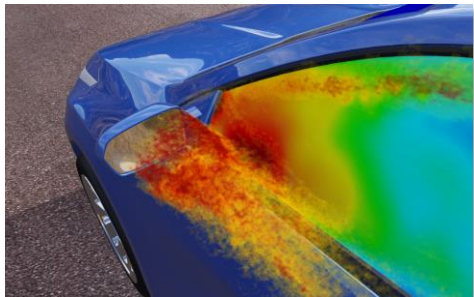
高保真几何和自动网格划分

Cut preparation time from weeks to days on fully detailed geometries



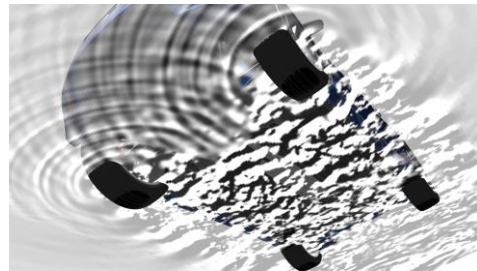
高效的求解器

10x faster than LES for comparable accuracy



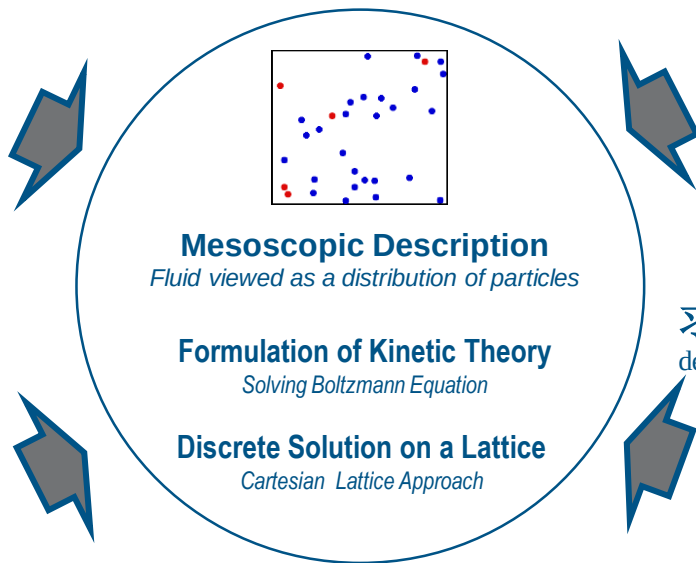
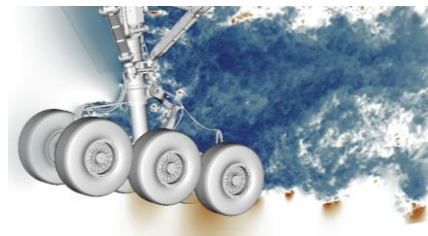
数值耗散低

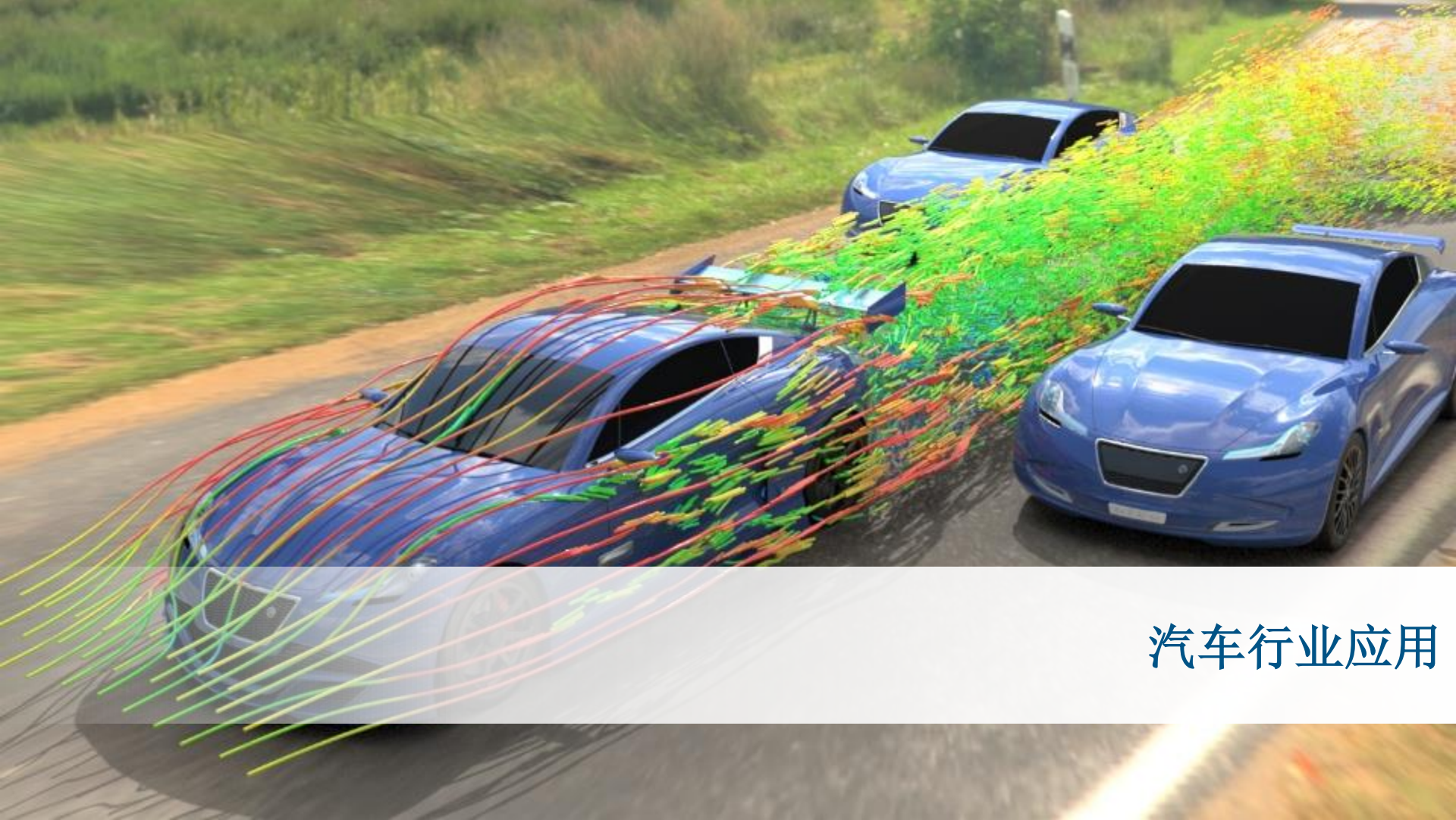
Accurate aeroacoustics near field direct predictions



求解复杂流动问题

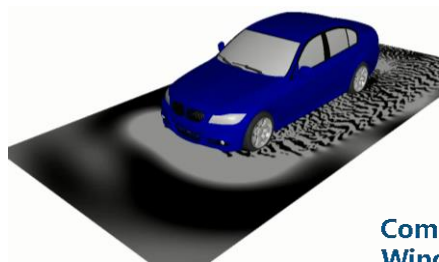
deal for full detailed geometries and off-design





汽车行业应用

PowerFLOW – 气动噪声



Underbody
Wind Noise
底盘噪声



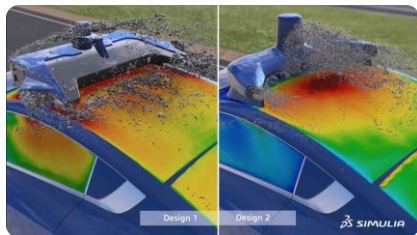
Greenhouse
Wind Noise
后视镜噪声



Component
Wind Noise
部件噪声



HVAC Noise
空调噪声



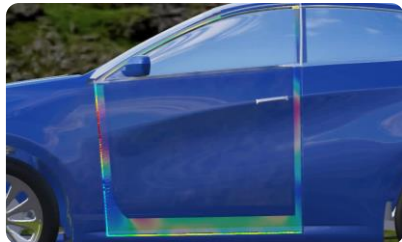
Seal Noise
密封噪声



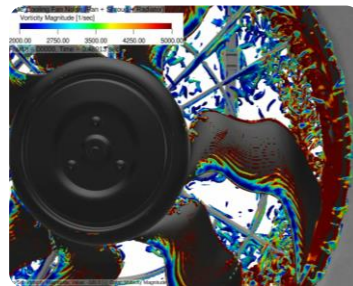
Sunroof/Side
Window Buffeting
天窗/侧窗风振



Cooling Fan
Noise
冷却风扇噪声



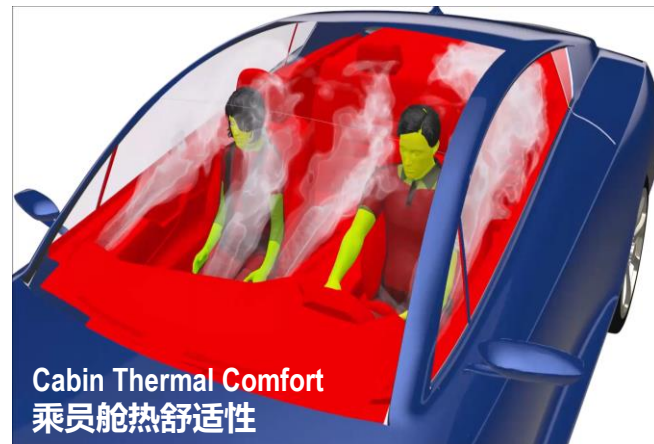
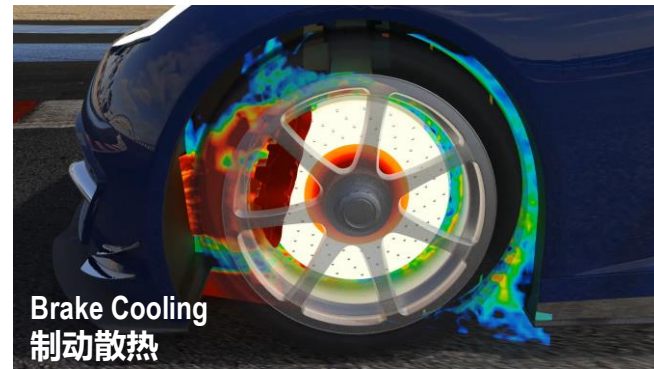
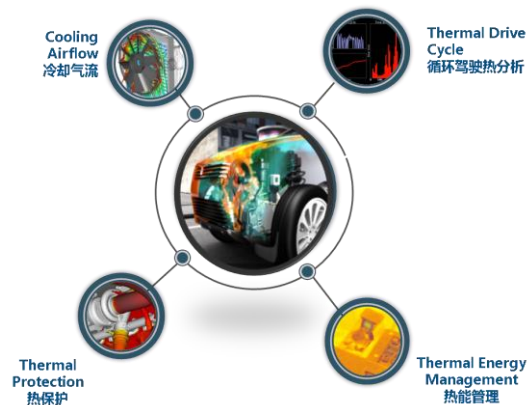
Simulations Include Detailed HVAC Unit
Geometry with Complete Vehicle Cabin Interior



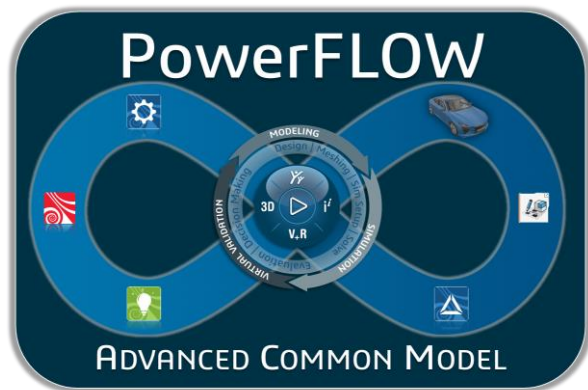
PowerFLOW – 空气动力学



PowerFLOW – 热管理



功能优势



ACM自动化 workflow，支持气动、风噪、污染与水管理等

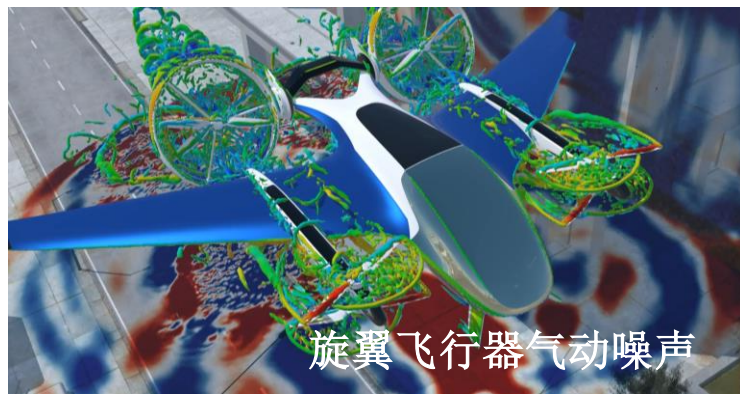
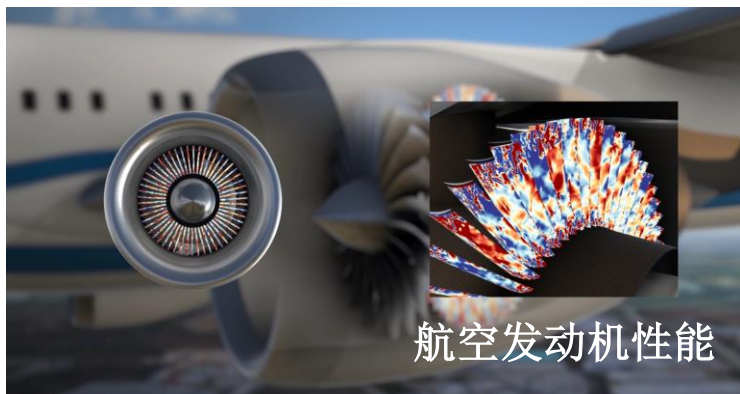
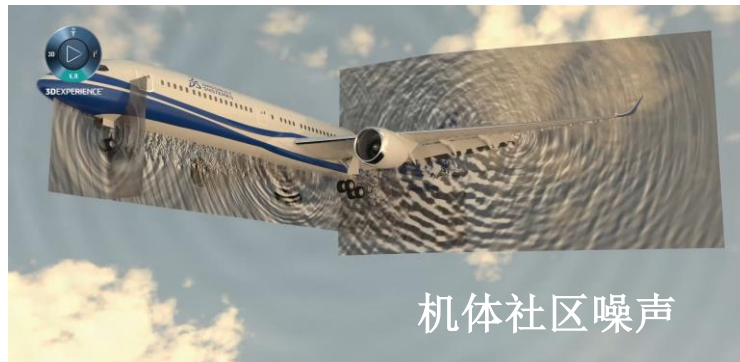
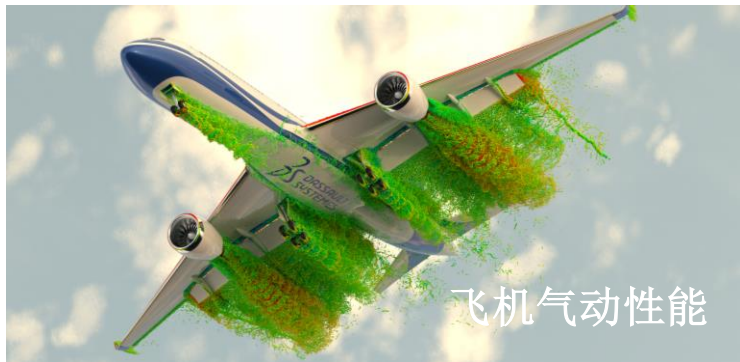
专业的空气动力学、气动声学、热管理解决方案

行业领先的精度验证，缓解风洞资源压力，降低成本，加速新车上市

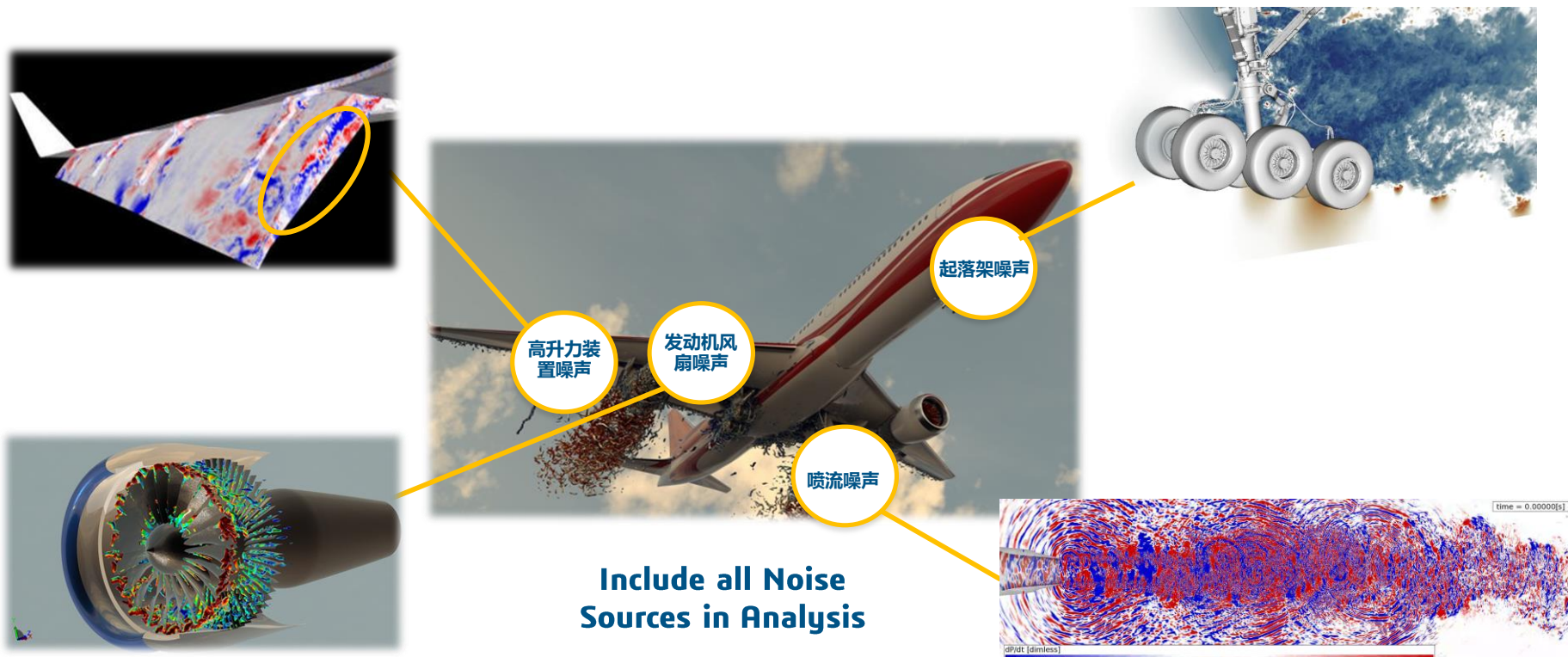
航空航天应用



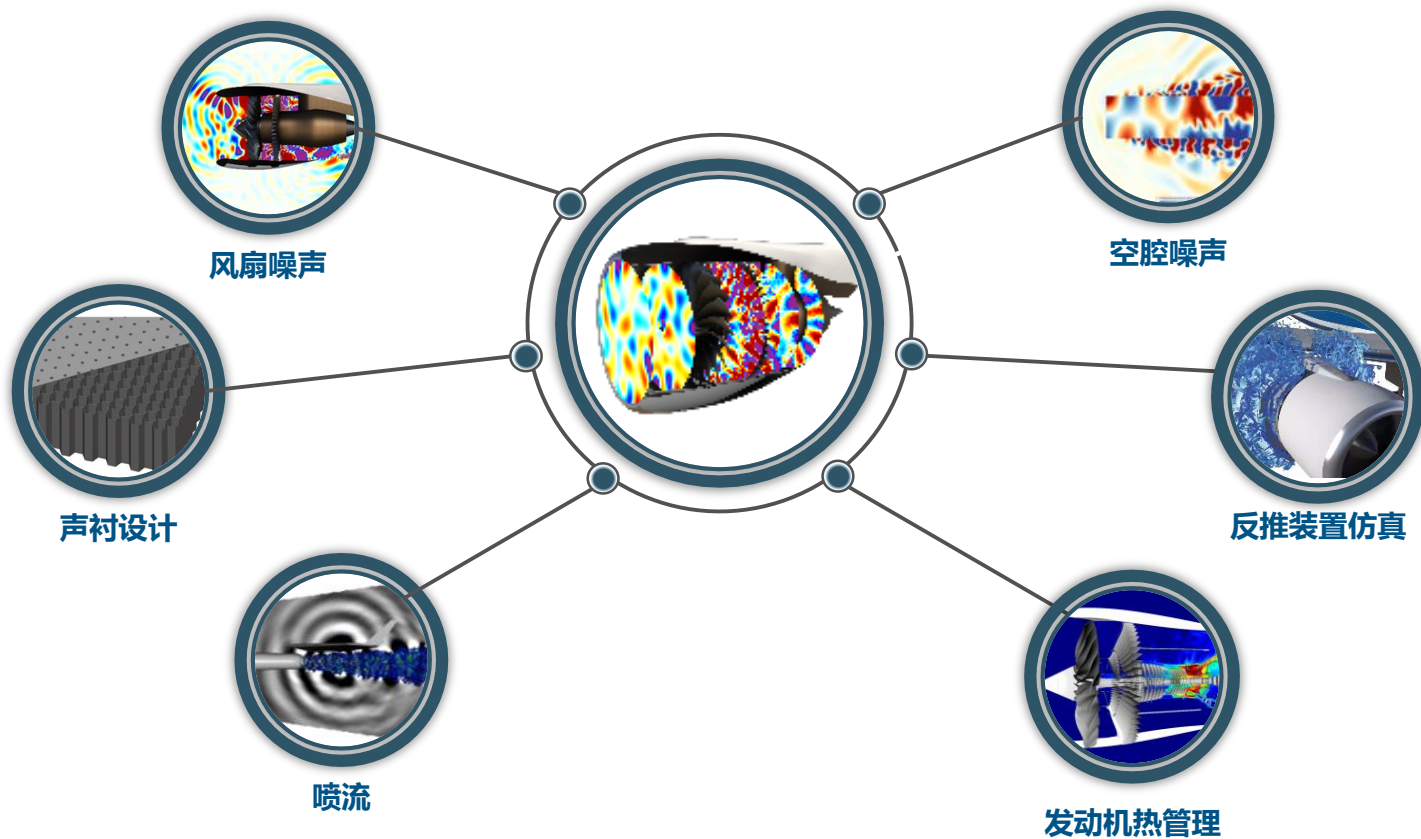
航空航天应用



机体社区噪声



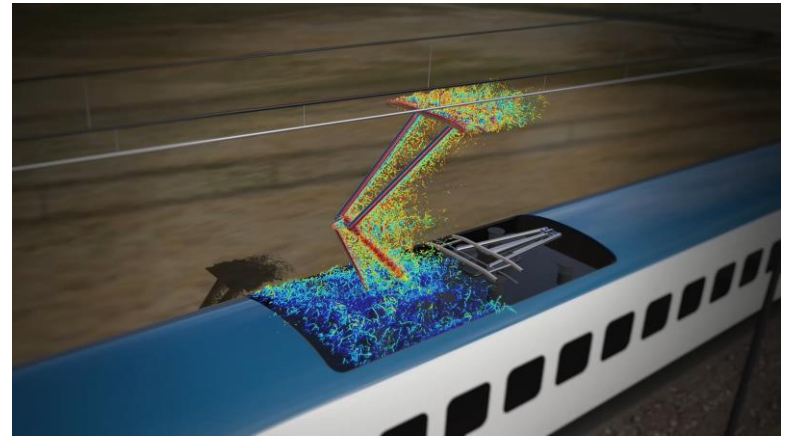
航空发动机性能



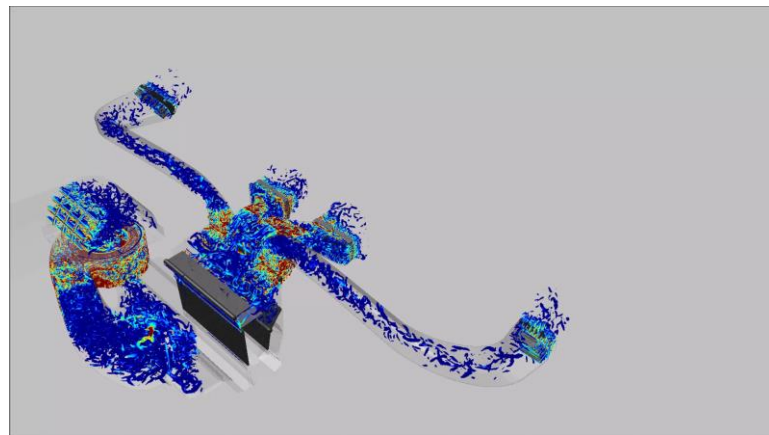
其他行业应用



Windfarm dBA max	dBA/turbine	#. of turbines
50	44	4
50	43	5
1dBA reduction = 25% more AEP		



其他行业应用



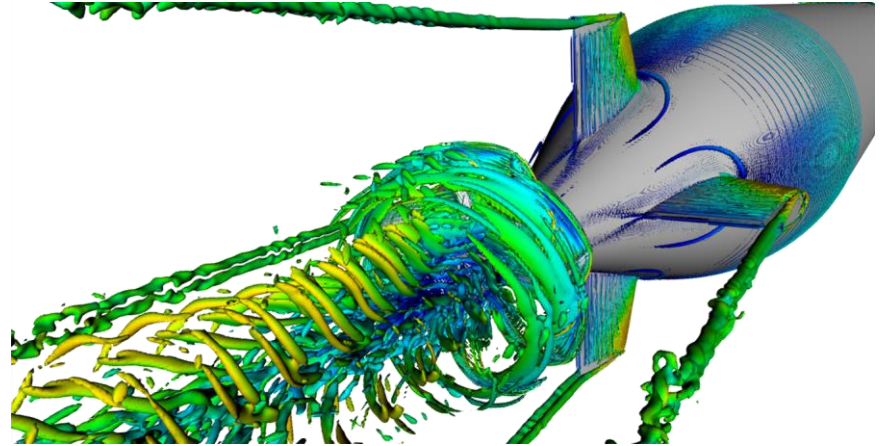
其他行业应用

Submarine Hydrodynamics

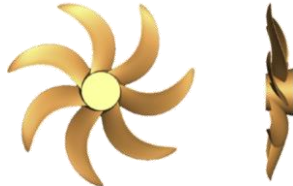
- Submarine model DARPA Suboff
 - AFF-1 without appendages
 - AFF-8 with appendages



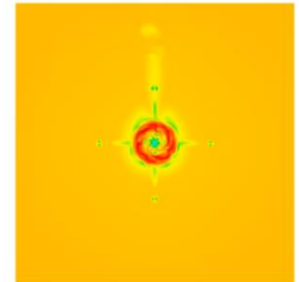
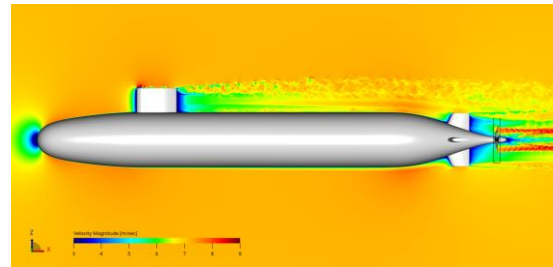
Geometric Characteristics of DARPA (Defense Advanced Research Projects Agency)
SUBOFF Models (DTRC Model Numbers 5470 and 5471)
<https://apps.dtic.mil/sti/citations/ADA210642>



- Propulsion unit INSEAN E1619
 - Open water, isolated conditions
 - Self propulsion, installed conditions



Geometry and data from Institute of Marine Engineering (CNR-INM, formerly CNR-INSEAN)



AGENDA

SIMULIA流体产品总览

PowerFLOW

XFlow、FMK、VPML

2025合作伙伴培训计划



XFLOW

专注于搅油润滑的CFD软件



3DEXPERIENCE®



XFLOW || 核心功能



高保真单相&多相求解器

- ▷ 基于粒子的格子玻尔兹曼瞬态求解器
- ▷ WMLES湍流模型



高效GPU加速

- ▷ 兼容NVIDIA和AMD显卡



易用性

- ▷ 无需手动划分网格
- ▷ 设置简便
- ▷ 轻松处理复杂几何形状



自然处理运动问题

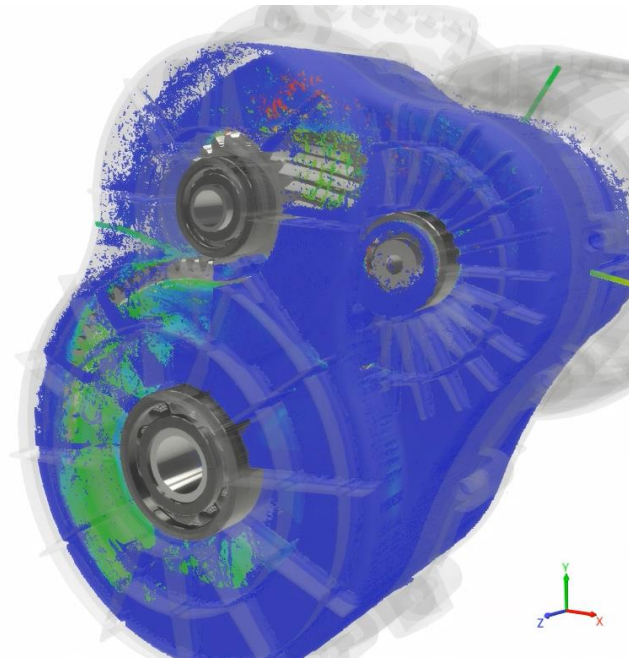
- ▷ 无需MRF或其他运动近似技术



Power'By

- ▷ 与3DEXPERIENCE平台连接

Time: 2.006s



XFlow



XFLOW GPU || 齿轮箱润滑

用户体验

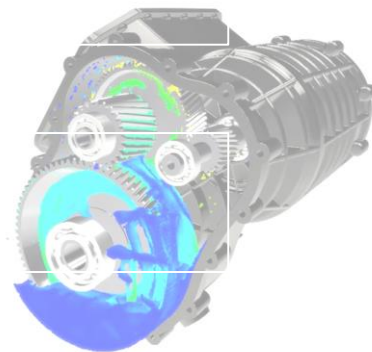
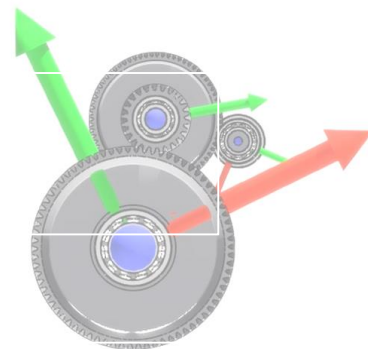
直观的图形用户界面，简化案例设置与可视化
通过脚本实现高效自动化，自动提取关键性能指标KPIs

性能

专为GPU优化的求解器架构

物理机理

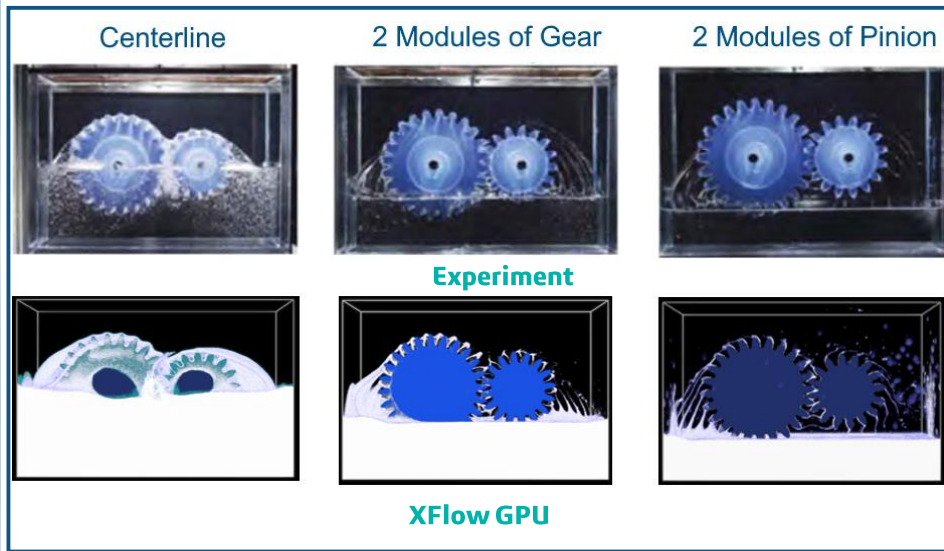
经过学术与工程验证的精确动力总成应用
全面的热分析能力



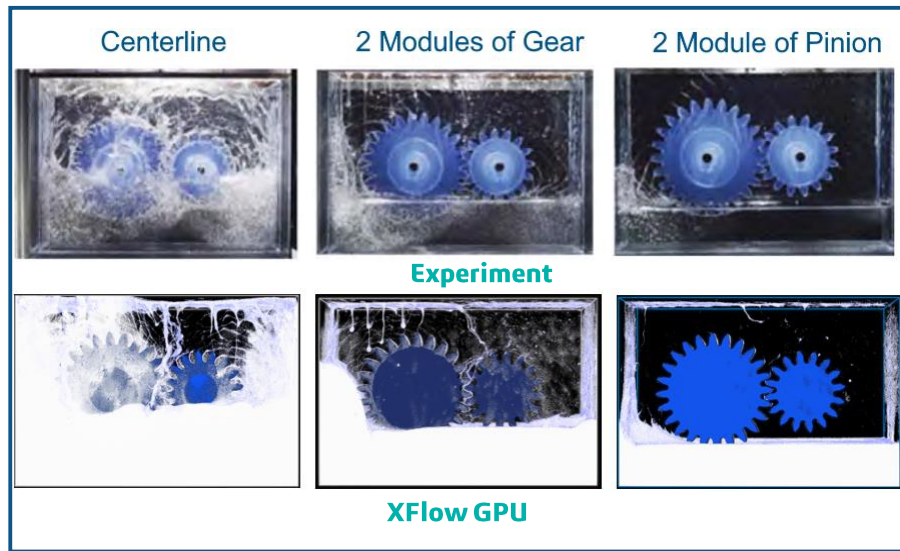
定性实验验证 || 油液飞溅模式*

- XFlow准确地复现了实验中齿轮在不同初始油位和切向速度下的飞溅模式。

Pitch speed $V_t = 0.88$ m/s



Pitch speed $V_t = 2.64$ m/s

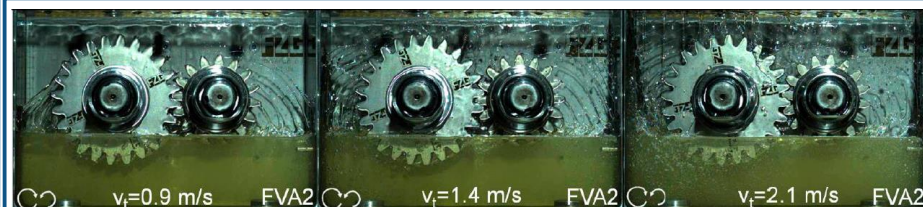


*Experiment reference:

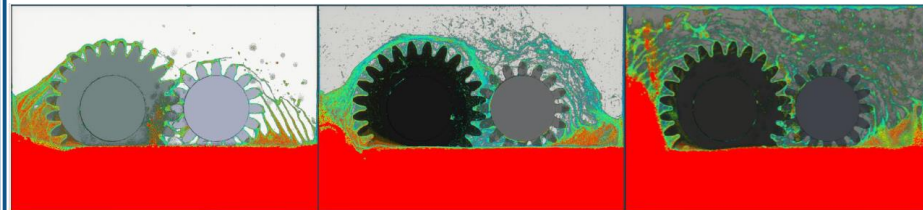
定性实验验证 || 油液飞溅模式*

- XFlow准确复现了相同初始油位，不同油液粘度和切向速度的油液飞溅情况。

Oil kinematic viscosity(@40°C)= 32mm²/s

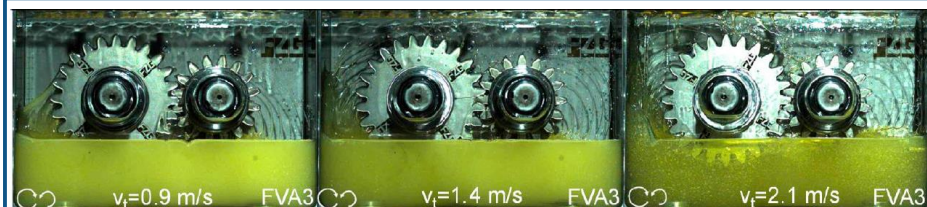


Experiment

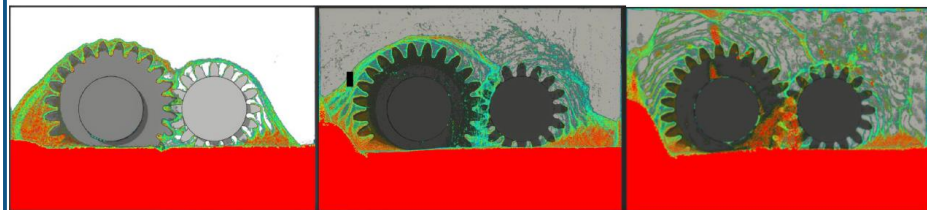


XFlow GPU

Oil kinematic viscosity(@40°C)= 95mm²/s



Experiment

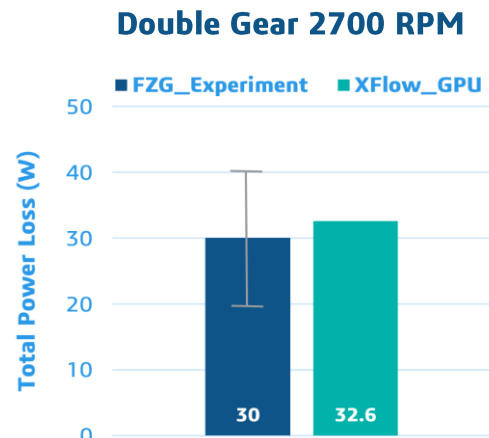
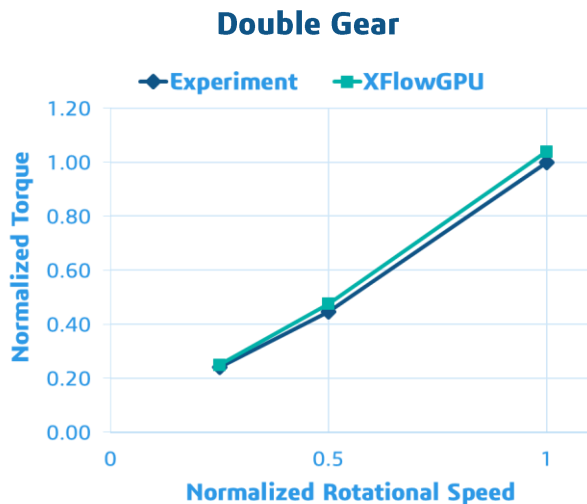
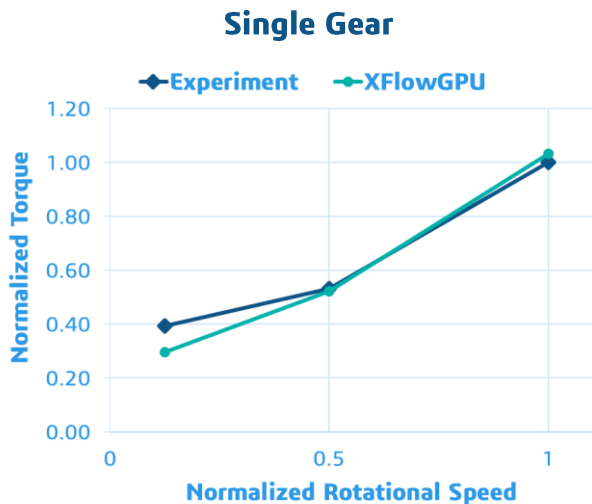


XFlow GPU

*Experiment reference:

定量实验验证 || 扭矩和搅油损失

- XFlow可以准确预测扭矩与搅油损失



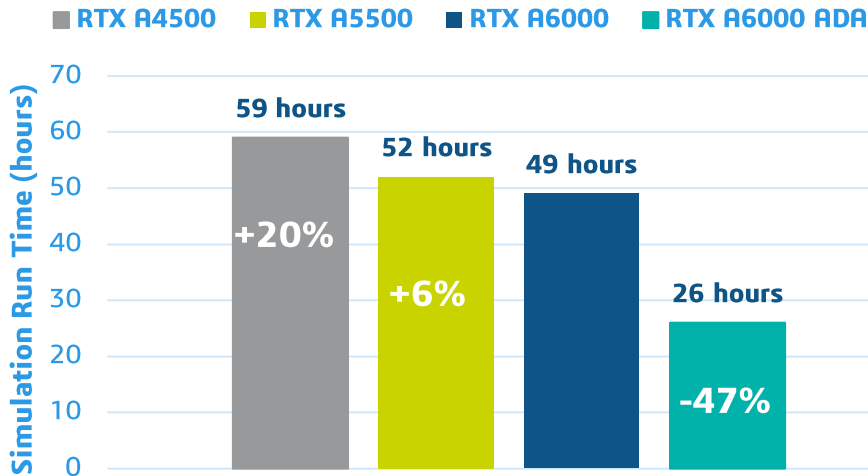
*Experiment reference:

Detailed Investigations on the Oil Flow in Dip-Lubricated Gearboxes by the Finite Volume CFD Method, Liu, H.; Jurkschat, T.; Lohner, T.; Stahl, K., Lubricants 2018

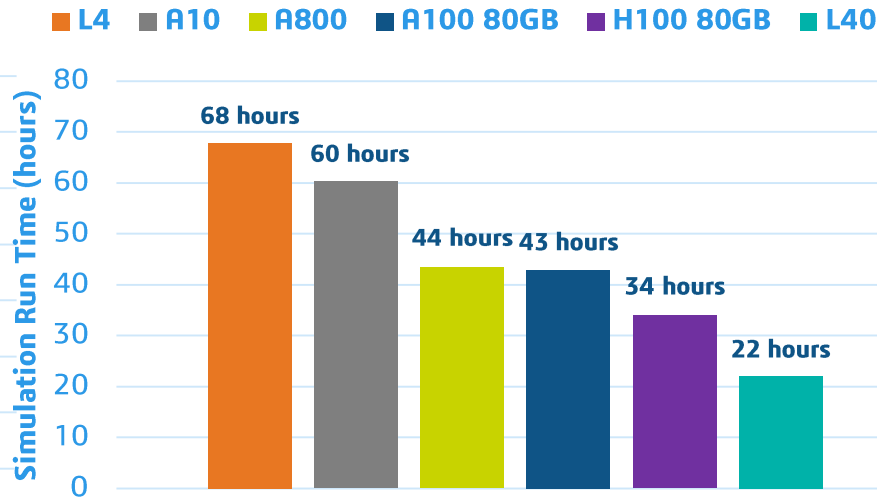
GPU 计算性能

- ▶ 本地工作站亦可实施，降低了设备采购和维护成本。

工作站GPU



服务器GPU



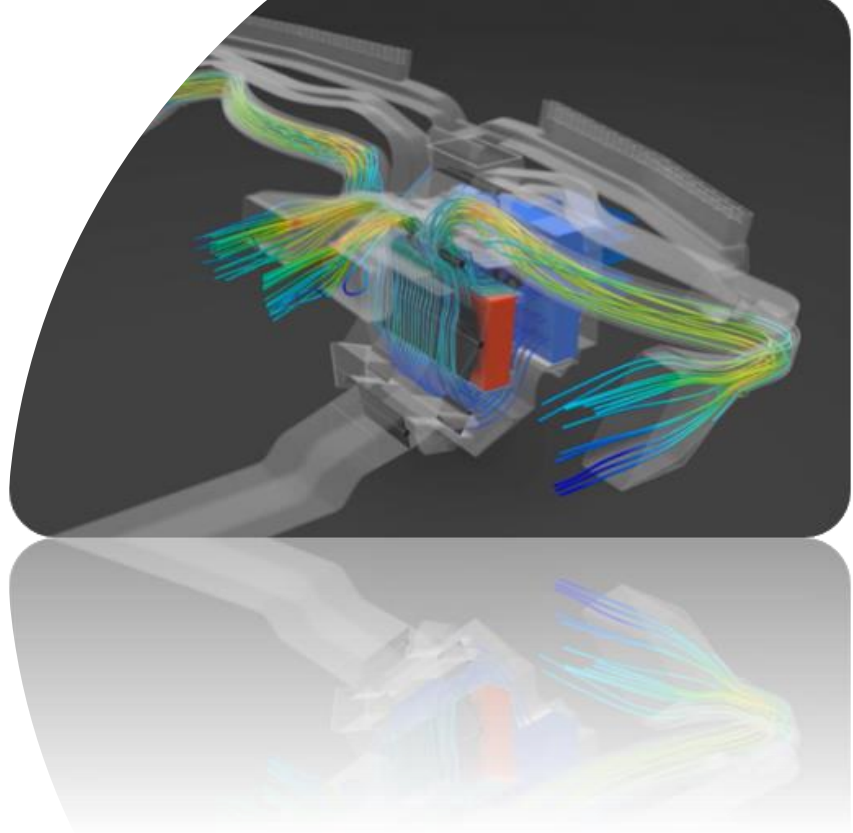


3DEXPERIENCE®

FLUID DYNAMIC ENGINEER

3DEXPERIENCE平台上的主流CFD

DS DASSAULT
SYSTEMES | The 3DEXPERIENCE® Company



全面性

稳态与非稳态流动

层流、湍流

MRF、动网格

颗粒、自由液面、多组分、多孔介质、辐射

Fluid Dynamics Engineer



本地/云计算

内置16核计算能力
可在本地和云端访问更多核心
快速、可扩展的求解器

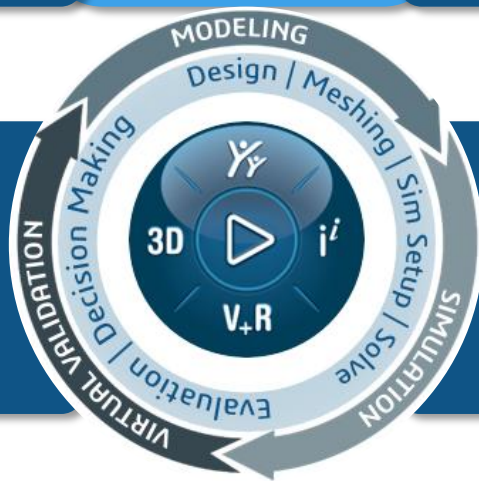
MODSIM 建模仿真一体化

与CAD和PLM的全面集成

与SOLIDWORKS和CATIA的无缝操作

——自动模型更新

3DEXPERIENCE平台上的协作工具

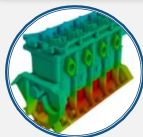


设计师为中心

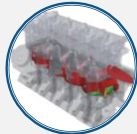
引导式仿真工作流程
从零件中轻松提取流体区域
用于设计探索的PDS App



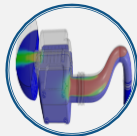
Powertrain Thermal Management



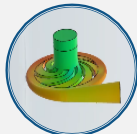
Engine Water Jacket Cooling Performance



Coolant Fill & Flush

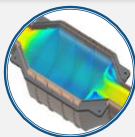


Air Intake Pressure Drop



Centrifugal Pump Flow Performance

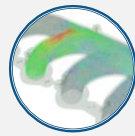
Exhaust System Engineering



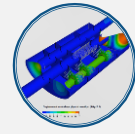
Catalytic Converter Flow Performance



Exhaust Manifold Flow Performance



Exhaust Manifold Thermal Cycling



Exhaust System Drain Performance

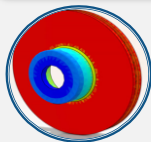
Cabin Comfort



HVAC Flow Performance

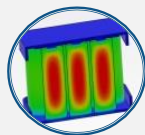


Brake System Engineering



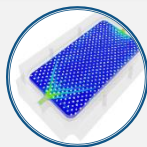
Component Brake
Cool Down

Battery Module & Pack Engineering



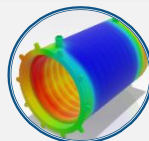
Battery Module
Thermal Management

Power Electronics Engineering



Thermal Management

Electric Drive Engineering



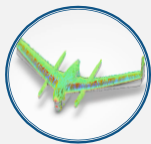
Jacket Thermal
Management



Aircraft Aerodynamics



Concept Aircraft
Aerodynamics

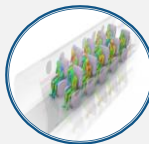


Concept Aircraft
Aerodynamics FSI

Aircraft Cabin Comfort

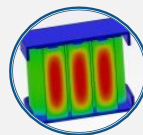


Passenger
Thermal Comfort



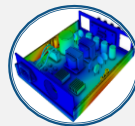
Seating Layout
Exploration

Battery Module & Pack Engineering



Battery Module
Thermal Management

Electronics Thermal Performance

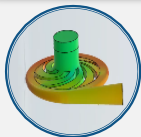


Electronics Enclosure
Thermal Management

目标行业 workflows | IE



Power & Fluidic Equipment Engineering

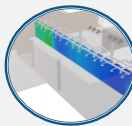


Centrifugal Pump
Flow Performance



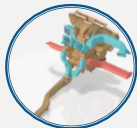
Valve Design

Built Environment Performance Analysis



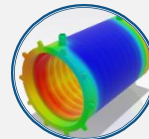
HVAC Systems
Integration

Heavy Machinery Safety & Comfort



HVAC Flow
Performance

Heavy Machinery Electrification Engineering

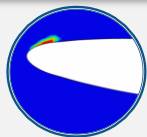


Jacket
Thermal Management

其他行业 workflows



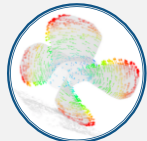
Marine Concept Performance



Cavitation Simulation



Hydrodynamic Analysis



Propeller Performance Analysis



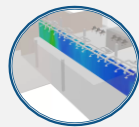
Packaging Qualification & Optimization



Flexible Packaging Analysis



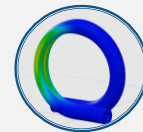
Built Environment Performance Analysis



HVAC Systems Integration



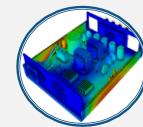
Device Mechanical Engineering



Pump Performance



Device Thermal Management



Electronics Enclosure Thermal Management

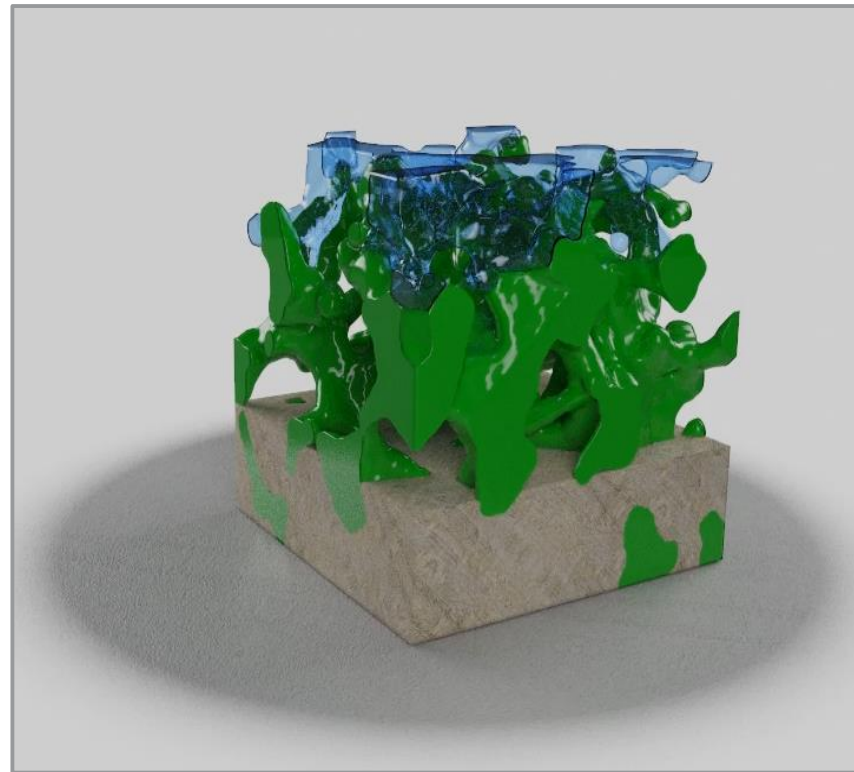


3DEXPERIENCE®

VIRTUAL POROUS MEDIA LAB

三维多孔介质多相流仿真

DS DASSAULT
SYSTEMES | The 3DEXPERIENCE® Company



VPML是3DEXPERIENCE平台内的一款应用程序, 旨在实现以下功能:

- ✓ 创建自然或工程多孔介质的数字孪生。
- ✓ 进行形态学分析。
- ✓ 执行单相及多相流体流动特性分析。

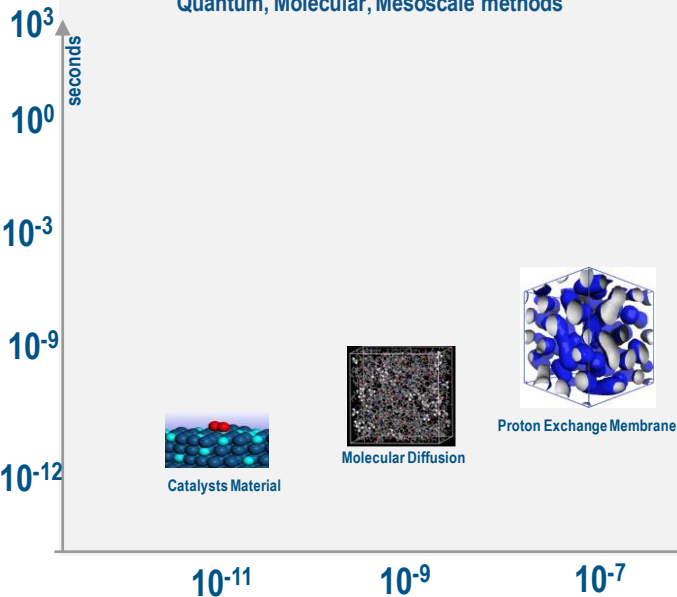
Virtual Porous Media Lab



优势:

- ✓ 实现分子尺度与连续介质尺度分析之间的数字连续性。
- ✓ 通过补充实验分析, 加速微观结构表征。
- ✓ 在无法进行物理分析的场所, 执行what-if分析。
- ✓ 缩短周转时间, 支持更多研究, 减少不确定性。
- ✓ 确定问题的根本原因, 并优化制造流程。

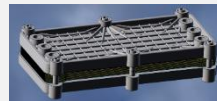
Quantum, Molecular, Mesoscale methods



Finite Element Method



Bipolar Plate



Fuel Cell Stack

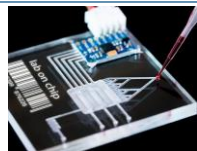
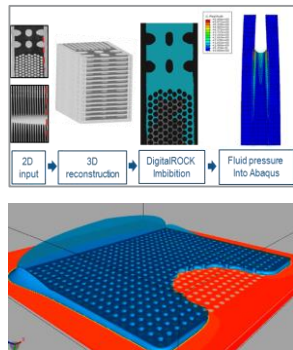
System Simulation Method



Vehicle

meters

VIRTUAL POROUS MEDIA LAB – 跨行业应用案例



微流控技术



Life Sciences
& Healthcare

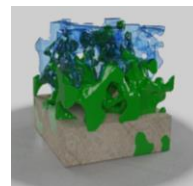


High-Tech

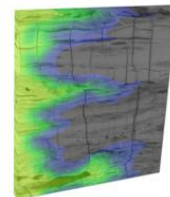
半导体



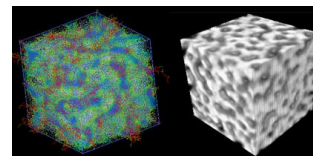
Infrastructure,
Energy
& Materials



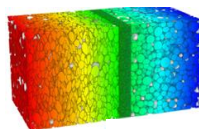
油气



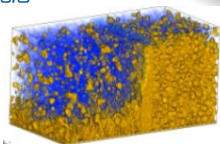
采矿



分子三维模型



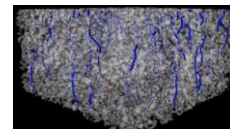
锂电池



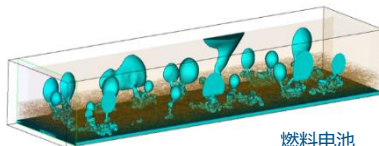
Transportation
& Mobility



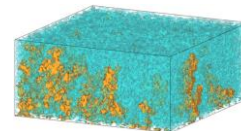
Industrial
Equipment



过滤器



燃料电池



电解槽

VPML用户见证



“some of the most important data is also the hardest to obtain requiring specialized drilling techniques and complex laboratory experiments that can take **a year or more**. We’ve taken our world-class fluid flow simulator and used it to power a digital version of a lab experiments providing central rock properties in **a matter of days**. This huge speed-up allows for much more analysis and deeper analysis”

AGENDA

SIMULIA流体产品总览

PowerFLOW

XFlow、FMK、VPML

2025合作伙伴培训计划

2025 SIMULIA FLUID TRAINING PLAN

培训主题	培训目标	课程分类	培训对象	培训讲师	方式	时长	时间
PowerFLOW基础培训	了解PowerFLOW技术特点、应用优势	初级	Partner Sales + Partner Tech	Changan Bai	Classroom	2days	25Q2
PowerFLOW基础培训	了解PowerFLOW技术特点、应用优势	初级	Partner Sales + Partner Tech	Grace Shen	Classroom	2days	25Q4
PowerFLOW初级： PowerFLOW介绍	了解PowerFLOW技术特点、应用优势	初级	Partner Sales + Partner Tech	Changan Bai	Webinar	1 hour	25Q3
PowerFLOW初级： PowerDELTA简介和软件操作	掌握PowerDELTA基础操作	初级	Partner Tech	Yuqin Deng	Webinar	1 hour	25Q3
PowerFLOW初级： PowerCASE简介与软件操作	掌握PowerCASE基础操作	初级	Partner Tech	Meng Liu	Webinar	1 hour	25Q3
PowerFLOW初级： 软件安装与SR提交方法	掌握PowerFLOW客户端和服务器安装方法，售后问题提交与跟进方式	初级	Partner Tech	Guanjun Wang	Webinar	1 hour	25Q3
PowerFLOW高级： 后视镜风噪仿真流程	1. 掌握PowerFLOW后视镜风噪仿真流程 2. 可独立完成算例设置与后处理 3. 可以给客户提供操作培训	高级	Partner Tech	Grace Shen	Webinar	1 hour	25Q4
PowerFLOW年度关注： 整车热管理解决方案	了解PowerFLOW整车热管理方案优势	年度关注	Partner Tech	Yang Yang	Webinar	1 hour	25Q4
PowerFLOW年度关注： ACMB介绍	了解ACMB自动化流程方案和优势	年度关注	Partner Sales + Partner Tech	Xiaowei Zhao	Webinar	1 hour	25Q3

2025 SIMULIA FLUID TRAINING PLAN

培训主题	培训目标	课程分类	培训对象	培训讲师	方式	时长	时间
Xflow基础培训	了解XFlow技术特点、应用优势	初级	Partner Sales + Partner Tech	Daisy Dai	Classroom	1days	25Q2
Xflow基础培训	了解XFlow技术特点、应用优势	初级	Partner Sales + Partner Tech	Daisy Dai	Classroom	1days	25Q4
XFlow初级: XFlow介绍	了解XFlow技术特点、应用优势	初级	Partner Sales + Partner Tech	Daisy Dai	Webinar	1 hour	25Q3
XFlow高级: XFlow润滑仿真	1. 掌握XFlow 齿轮箱润滑仿真流程 2. 可独立完成算例设置与后处理 3. 可以给客户提供操作培训	高级	Partner Tech	Daisy Dai	Webinar	2 hour	25Q4
FMK 基础培训	了解FMK技术特点、应用优势	初级	Partner Sales + Partner Tech	Grace Shen	Classroom	1days	25Q2
FMK初级: FMK介绍	了解FMK技术特点、应用优势	初级	Partner Sales + Partner Tech	Dmitriy LIKHACHEV	Webinar	1 hour	25Q3
FMK高级: FMK HVAC仿真	1. 掌握FMK HVAC仿真流程 2. 可独立完成算例设置与后处理 3. 可以给客户提供操作培训	高级	Partner Tech	Dmitriy LIKHACHEV	Webinar	2 hour	25Q4



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